

Jio Subscriber Churn Intelligence

Executive EDA Summary — Workstream 1 Diagnostics

Executive Summary

This report distills the Workstream 1 diagnostic analysis of 64,738 Jio subscribers into an executive brief. The current 30-day churn rate is **1.73%**, concentrated most heavily in the 'Uttar Pradesh East' circle (2.8% churn). A clear 'silent churn' early-warning signal emerges once the recharge gap reaches the '90+'-day band. Pricing sensitivity and network quality-of-service both show measurable relationships with churn, giving the business three concrete levers - geography, recharge timing, and price/quality perception - to act on immediately.

Context: Subscriber Churn & Revenue Exposure

Even a small churn rate translates into material revenue exposure at Jio's subscriber scale. The table below frames the headline diagnostics in business terms to support prioritization discussions ahead of the Workstream 2 predictive-modeling build.

Metric	Value
Total Subscribers Analyzed	64,738
Baseline 30-Day Churn Rate	1.73%
Churned Subscribers (30d)	1,122
Average ARPU (Last Month)	INR 213.70
Est. Monthly Revenue at Risk	INR 239,771
Top-Risk Circle	Uttar Pradesh East (2.8%)
Silent-Churn Recharge Gap Threshold	90+ days

Note: revenue at risk is an illustrative estimate (churned subscribers × average ARPU) and excludes indirect effects such as referral or bundle attrition.

Visual Section 1: Geographic Churn Hotspots & Recharge Silent Threshold

1.1 Circle-Level Churn Hotspots

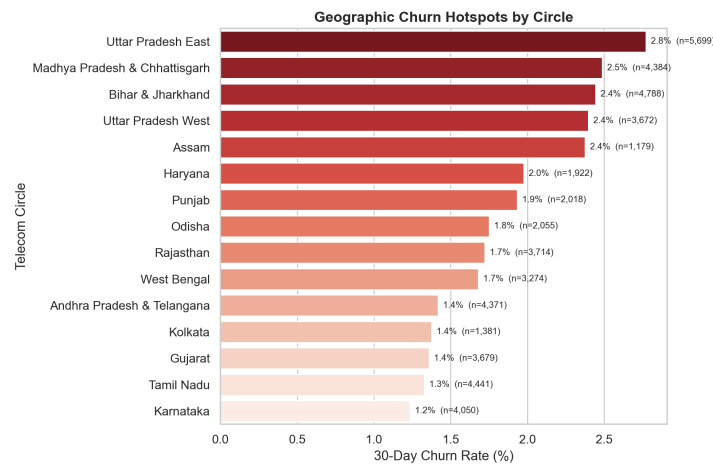


Figure 1: 30-day churn rate by telecom circle, ranked highest to lowest.

- 'Uttar Pradesh East' is the highest-risk circle at 2.8% 30-day churn, vs. a network-wide average of 1.6% across all circles.
- 10 of 22 circles sit above the network average churn rate - prioritize retention budget in the top 3-5 circles shown above for the fastest ROI.

1.2 Recharge Gap & Silent Churn Threshold

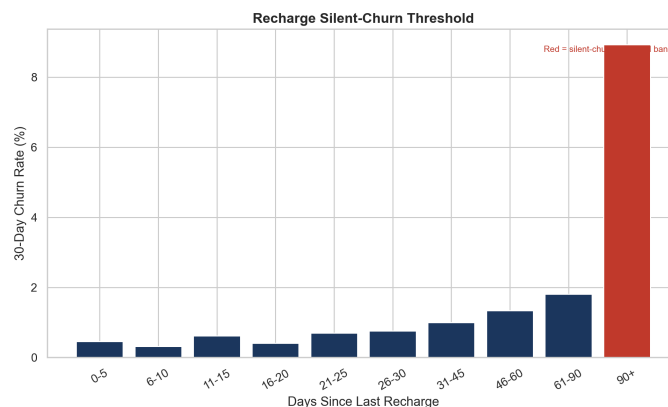


Figure 2: 30-day churn rate by days since last recharge; the silent-churn threshold band is highlighted in red.

- Churn risk accelerates sharply at the '90+'-day recharge gap band (8.9% churn), the clearest 'silent churn' early-warning signal.
- Subscribers recharging within the first 5 days of their cycle show only 0.5% churn - proactive outreach should trigger before this threshold.

Visual Section 2: ARPU Price Sensitivity & Network QoS Correlation

2.1 ARPU Price Sensitivity

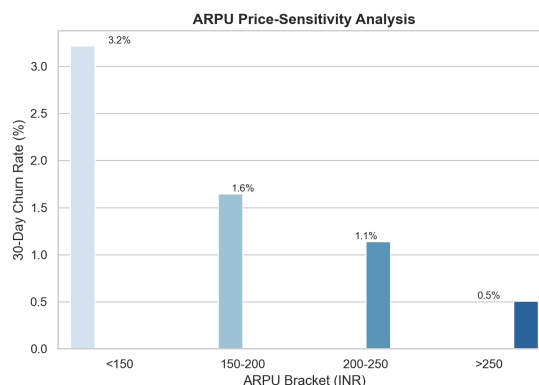


Figure 3: 30-day churn rate by ARPU bracket (INR).

- The '<150' INR ARPU bracket shows the highest churn rate at 3.2%, signaling a value-for-money gap in that price tier.
- The '>250' INR bracket is the most stable at 0.5% churn - a useful benchmark for tiered retention offers calibrated by ARPU bracket.

2.2 Network Quality-of-Service Correlation

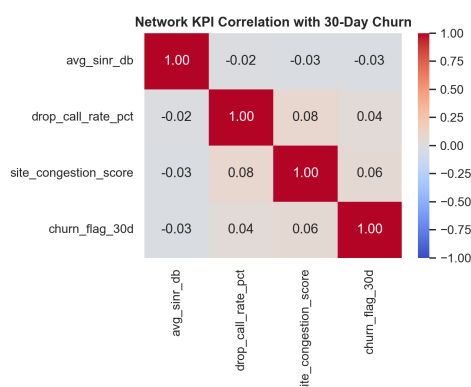


Figure 4: Correlation matrix of network KPIs with 30-day churn.

- 'site_congestion_score' has the strongest relationship with churn ($r = 0.06$); as it rises, churn risk increases, confirming network experience as a material churn driver.
- Network KPI correlations are directional signals, not standalone predictors - they should be combined with behavioral and pricing features in the Workstream 2 model.

Strategic Next Steps — Workstream 2 (Predictive Modeling)

- Engineer a recharge-gap feature set anchored on the silent-churn threshold identified here, including days-since-last-recharge and recharge cadence trend.
- Include ARPU tier and ARPU trend (3-month vs. 6-month average) as pricing-sensitivity features.
- Incorporate network KPIs (SINR, drop-call rate, congestion score) as service-quality features.
- Add circle/geography as a categorical feature or apply geographic stratified sampling, given the concentration of risk observed here.
- Apply class-imbalance-aware modeling techniques (class weighting, threshold tuning, or resampling) given the low baseline churn rate, and validate using precision-at-top-decile and recall rather than raw accuracy.